

World Class Engineering Student Final Paper

Goal Setting: While it may sound like stating the obvious, goal setting is the first step in achieving our goals. We may have an idea of what we want, but without a solid understanding of what the end result looks like, and the path we intend to get there, it can be difficult to stay focused on our goals, or to achieve them at all. A world class engineering student would understand the importance of goal setting and would implement a well thought plan to reach their goals. Their goals would be specific, like what degree they would like to obtain, when they want to graduate and with what academic standings such as honors or membership in a specific engineering society. After they had created specific goals that they would like to achieve, a detailed plan would be created to help guide them through their academic career. All this preparation is very important, but it would lack much of its value if the engineering student didn't have a good understanding of what engineering is and the many diverse positions out there, each with a unique role in society. Understanding our goals is motivating. It creates an excitement in us when we see what our hard work is leading to, and drives us to put our head down and keep working hard.

Prior to this class, I didn't have well defined goals let alone detailed plans on how to reach them. In hindsight, this has become very evident to me in my previous three years of college. When I graduated High School, I stepped into the world without much understanding of what college would look like for me, and why I was really going. I quickly crossed off a four year university from my list of options due to financial reasons, amongst a few other reasons that I probably could have worked around. The only real academic advice I had received and paid attention to, other than "go to college", was from my high school metal shop teacher. During the four years of shop, he worked to encourage me and my peers who had a deeper interest in machining to pursue a degree in manufacturing from Portland Community College (PCC). Having discovered that I greatly enjoyed machining I decided that I would go for an Associates of Manufacturing Technology from PCC. I knew my goal was to get my associates, but that was about as specific as it got. I had little understanding what a career would look like with that degree, or even how to get the degree. I simply decided to enroll in college the first term after high school, and to take a math class every term until I was finished with the required math

courses. I did meet with an academic advisor, but only once. It was required to meet with an advisor during the enrollment process, but that was the only interaction I had with the college staff, other than professors during class. After three years with no real direction, and dwindling motivation due in part to the frustration that resulted from randomly taking courses, I found myself with only about 40 credits and a sense of failure.

Last year I was hired on as a machinist at a job shop and began training to further my skills. While I enjoyed (and still do) my work, I came to the realization that I want more from my career and with some research and advice decided that I wanted to become a mechanical engineer. I realized that it was time to make a serious commitment to my education if I wanted to get a degree that would help me obtain a career I would enjoy. I decided that I wanted a degree from Portland State University (PSU), and transferred to Clackamas Community College (CCC) with the intention of completing my freshman and sophomore years here before transferring to PSU. With my busy work schedule I decided that to get the ball rolling I would enroll in at least one class, and that ended up being Engineering 111. I had little understanding of what the engineering field looks like and how to navigate through the education journey before taking this class, and have learned a lot in the last few weeks. I am now in the process of creating detailed goals with thoughtful plans to achieve them. So far I have determined that I want to earn a Bachelors in Mechanical Engineering and that it is going to take me about six or seven years to complete going to school part time. I still need to plot out my “road map” for the courses I am going to take at CCC and when I am going to take them. This will help me stay focused and give me a better idea of my academic timeline. I need to continue to strengthen my commitment to my goal, but I do believe that I am taking it much more serious than my previous goal at PCC, which is very encouraging. I’ve noticed that the more I learn about what an engineer does, and the more details I learn about the education process for an engineering student, the more excited and determined I become. It is my goal to not only graduate, but to do it with a high GPA, preferably above 3.50 and I would like to graduate with honors and membership in a mechanical engineering society as well. I understand that this is going to be a difficult process that it will require lots of hard work and dedication; however I believe that I am ready and willing to make the necessary sacrifices that I can to complete my Bachelors.

Community Building: Being actively involved in a community has numerous benefits and I believe that as humans, this is where we thrive. A world class engineering student would recognize the importance and immense value of a community at school and would work towards becoming well established in it right from the beginning. They would build relationships with their peers in the courses they are taking, as well as the professors and other school faculty that they interact with. This would be apparent through them reaching out to peers to create study groups, lab partners and to genuinely get to know who they are in school with. It would be difficult for them not to create new friendships through this process. A world class engineering student would also be sure to keep in contact with their professors by communicating with them not only during lectures but also during office hours by seeking academic advice or help from them. While establishing relationships in the class room are extremely helpful, the student would also become involved with other aspects of campus life including activities like tutoring, engineering clubs and even student government. An example student may even pursue a part-time job on campus to help supplement the cost of tuition, as opposed to a job off-campus. This would benefit them by helping them to spend even more time on campus, thus continuing to engage in the community.

In all honesty, when it comes to the community at school I am not very involved. While I have only been at CCC for a few weeks, I didn't really do much to be involved during my time spent at PCC. I often won't go out of my way to speak to people in class, and rarely spend more time on campus than required to attend my courses. I have never been involved with any co-curricular activities or clubs of any sort.

If I want to be successful in school, and later on in my career, I need to learn how to engage in the community at school; and I've already began that process! After reading the text book we are going through, paired with our in class lectures, I've gained a better understanding of how important it is to have relationships with my peers and professors. I recognize that this is going to require a good amount of personal adjustment, and effort on my part, but I'm excited to work on it. The engineering design project has required me to work with a classmate, which I was nervous about at first, but have come to greatly enjoy. Together we made an effort to set aside a time each week outside of class to work together on our project and the productivity that has resulted is very satisfying. It is my goal to create new friendships and to know at least three

new people by name in each of my courses. I will strengthen these relationships by creating study groups with my peers whenever possible. I want to be intentional about knowing and addressing my professors by name, and making sure they know who I am. In addition to building relationships, I would like to schedule time for myself to become involved in some sort of co-curricular activity, or club of some sort while I am at CCC; however I'm not certain yet what that may be. This is a plan that I will need to be constantly working towards, but I understand that it will have immense benefits.

Academic Development: There are many resources available to students to help enhance their education, but as a student you have to take the initiative to utilize those resources. A world class engineering student would take the time to research what resources are available at their particular school. After identifying the tools that would help them the most, they would actually use them. Spending time with tutors and using the resources available in the school's library and computer labs would provide opportunities to reinforce the material they are learning, while engaging in the campus community. An example student would have a solid understanding of how their institution's engineering program is structured, in addition to school policies and student rights. Frequent interaction with academic advisors, perhaps meeting every term, would be very beneficial in planning a road map for the courses that need to be taken, and to stay on track with that plan. It is important that students intending to transfer from a community college to a university understand what that transferring process looks like and what will be expected of them prior to transferring.

While I've always known that resources such as tutoring and academic advising were available to me as a student, I've never really taken advantage of those resources or looked to see what else the college might have to offer. I can't say that I've ever had a good understanding of how the program I am involved in is structured either. I tend to avoid joining clubs or participating in extracurricular activities as well, and honestly can't picture myself in student government.

As I learn more about how to become a successful student, and recognize that there many opportunities for success, I am beginning to think about what the college may have that can help me go above and beyond in my studies. I am aware that I need to take the initiative, as most resources like this won't come to you. Right now I still need to actually seek out what resources

might be helpful to me. I know that the library has tutors and a computer lab available to me, and have already spent time there working on the design project. I would like to learn about what clubs and extracurricular activities exist on campus, and see if any catch my interest. I'm sure there are many other resources available, and I am looking forward to enhancing my experience while building the habit of seeking out useful information that will help me reach my goals.

Personal Development: In order to be a successful world class engineering student, and further enjoy your educational experience in the process, you need to be relatively self-aware. An example student would understand that learning how to slow down and analyze where they are at in various areas of personal development would be a tremendous help in succeeding in school and in the rest of life. Students who practice and work at being "good students" generally do noticeably better in school, and tend to get more out of their education. The key is to understand that the learning isn't just in the classroom, at that they way you live outside of school has a direct impact on your education. Your time management skills, study habits, physical and mental health as well as your attitudes and behaviors all have an impact and can all be continuously improved.

This particular area is where I need, and want, the most growth. Two of my main pitfalls in school have been poor study habits and procrastination, which I've found really go hand in hand. Even in high school, I would avoid homework and certainly wouldn't study unless I had a project due the next day, and even then I occasionally decided to spend my time doing something else. For the most part I understood what was being taught to me, and usually scored very high on tests. I point that out only to compare it to my lack of consistently high grades to support those test scores. There were many classes that I could have received an A grade in but ended up with a C or even a D. I am aware that this is primarily due to my terrible habit of procrastination. During my last few years at PCC, my study skills weren't that much better. I still put things off, and with the increasingly difficult course work, I found myself growing more and more frustrated at the decreasing quality of my work. Even looking back now, I am disappointed in the low grades I received in many classes. Reading through our text book and reflecting on the many methods of personal development have been enlightening and very encouraging; in a way I wish I had come across some of this advice earlier on.

I do have a plan for growth, I've already began to implement it and I'm starting to see the benefits of doing so, which is very encouraging. When I transferred to Clackamas, I was aware that this was an opportunity for a start fresh, and I have every intention of doing well. I'm coming into this major with much more direction, understanding and motivation than I had for my previous major. Time management has always been a weak area of mine, and I've been working to improve it for many months now, with significantly more focus applied to how much time I spend studying. I am balancing full time work with school right now, and most likely will be for the majority of my schooling. I have acknowledged that time is valuable, and with my tight schedule, I can't afford to waste any. Throughout this class I have worked hard to set a high priority to my studies by starting to work on assignments as soon as possible so that I can space them out in manageable chunks. As a result I have been completing projects ahead of schedule and have found that I actually have more free time on the weekends, time that doesn't come with the stress of knowing I still have work to do. I am pretty satisfied with the momentum I have right now, and am continuing to work towards better organization and time management.

Staying in healthy physical and mental shape is continuing to become more important to me the older I become. I've been trying to run/jog regularly for almost two years now and that has been very encouraging. I've also found that when I exercise more frequently, I usually have more energy and am significantly more self-confident. This leads to me being more productive and more satisfied with the work I accomplish. I found that I can relieve a lot of stress by keeping my household clean and organized, which helps me decompress and comes with numerous other benefits. This term I have been practicing these and other methods of self-improvement and I really believe it is paying off. I am much more engaged in school and am more excited to finish my degree than I've ever been. The one major obstacle I am working on is networking with my peers. This is going to be outside of my comfort zone for sure, but it is necessary, and honestly I know I will enjoy it once I take the initiative with others to establish new relationships. Overall I think I am off to a good start in school now, and will be continuously looking for ways to improve and better myself so that I can be the best engineer I can possibly be, and truly enjoy it.